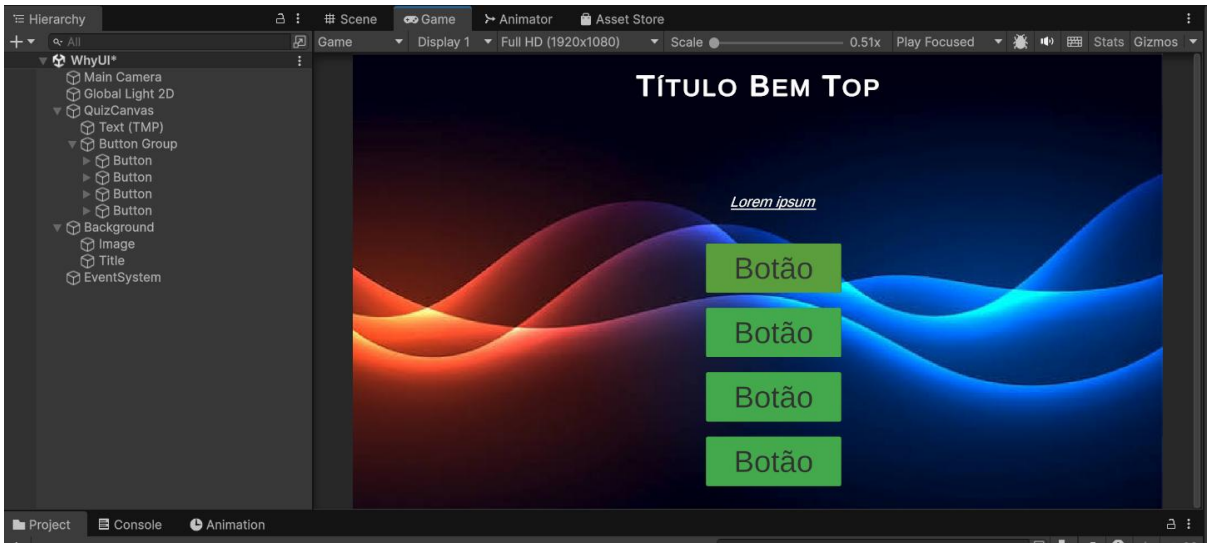
POLITECNICO SETUBAL

Technical Document 4 – UI and UX

Design e Programação de Jogos (DPJ)

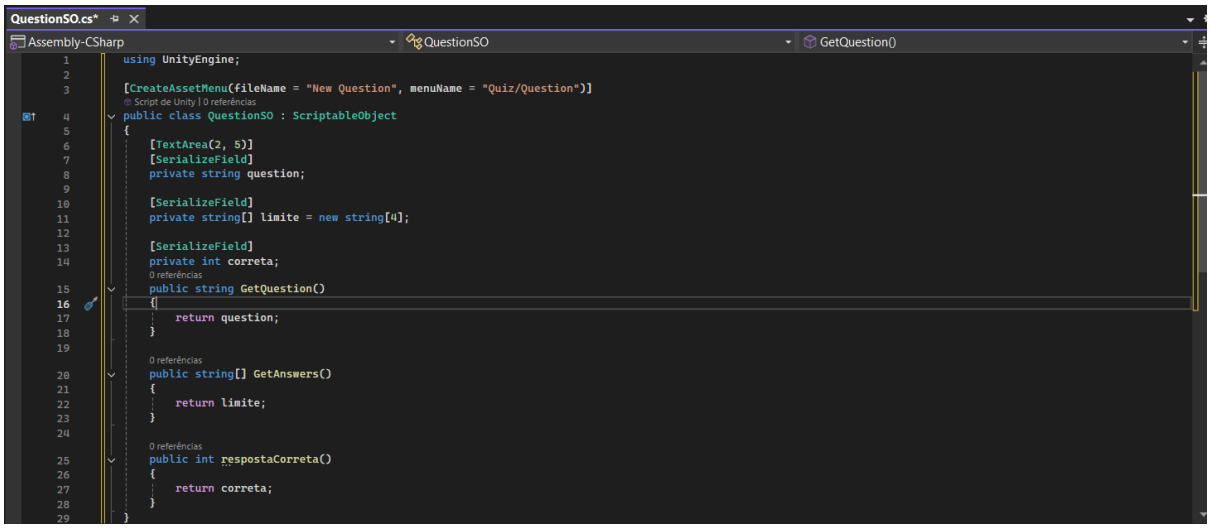
Level 1

On the first level we started making the layout of the UI and how we want it to be, choosing a background image, making two canvas, one for the buttons and the text of the question, and the other for the background



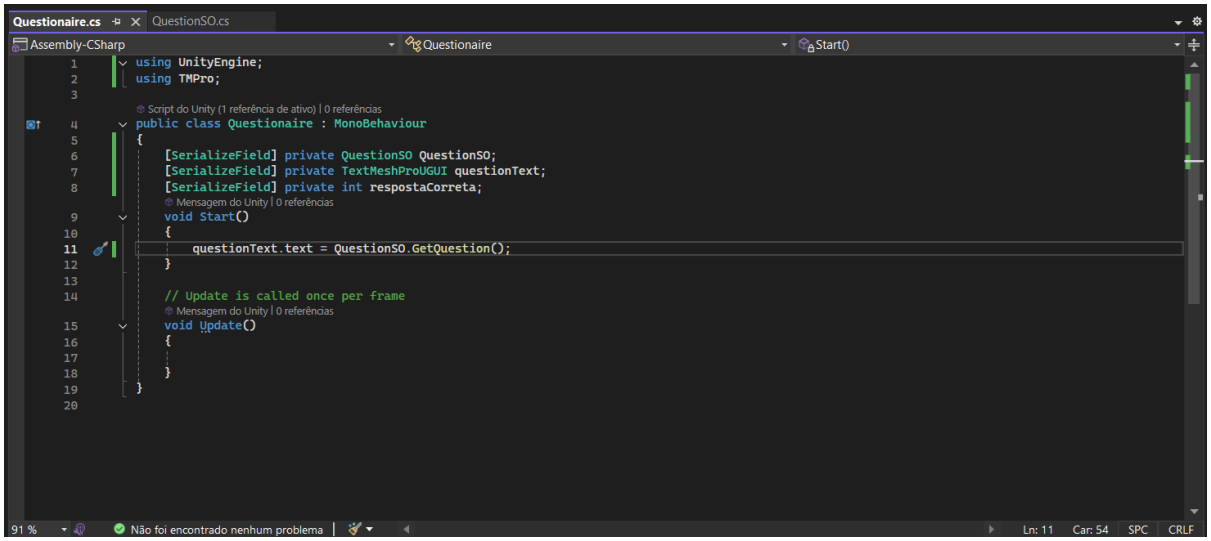
Level 2

In this level we made the first script of the Unity project, called QuestionSO that is linked to a Scriptable Object called QuestionSO, that will define what the question is, what the available answers are and what is the correct one.



Level 3

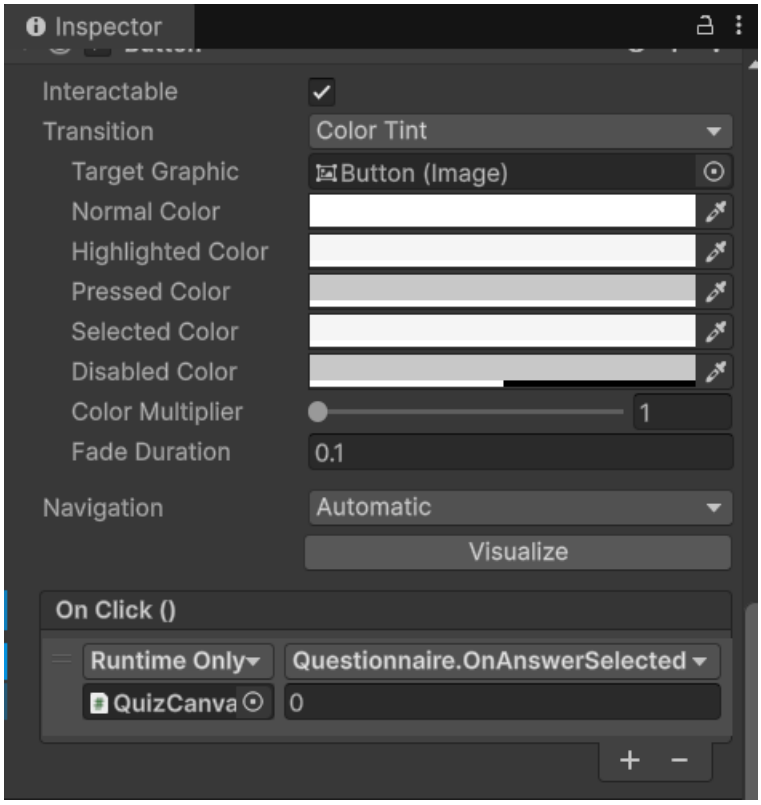
On the third level we made a new script called Questionaire, where we change the text of the question text created in the first level as you can see in the example on the second image of this level.





Level 4

On this level we created the On Click function for all the buttons, where when you click a button, it will go to the Questionaire script and use the OnAnswerSelected function.



```
0 referências
public void OnAnswerSelected(int index)
{

    if (index == correctAnswerIndex)
    {
        questionText.text = "Correto!";
    }
    else
    {

        string correctAnswer = questionSO.GetAnswer(correctAnswerIndex);
        questionText.text = "Desculpa, mas a resposta correta é: " + correctAnswer;
    }


    SetButtonState(false);
}
```

This is the OnAnswerSelected Function which tells us that if the index is equal to the correctAnswerIndex, it will change the question text to the text “Correto!” and if it isnt it will say “Desculpa, mas a resposta correta é:” and then the correctAnswer variable.

```
0 referências
public void SetButtonState(int state)
{

}


2 referências
public void SetButtonState(bool state)
{
    for (int i = 0; i < answerButtons.Length; i++)
    {
        Button btn = answerButtons[i].GetComponent<Button>();
        if (btn != null)
        {
            btn.interactable = state;
        }
    }
}
```

The main and most functional version uses a boolean parameter (bool state). Its purpose is to loop through every button stored inside the answerButtons array using a for loop. For each button in the array, it fetches the Button component and sets its interactable property to match the value of state.

Level 5



```
Script da Unity | 2 referências
public class Timer : MonoBehaviour
{
    [SerializeField] private float timeToCompleteQuestion = 30f;
    [SerializeField] private float timeToShowCorrectAnswer = 5f;

    public float timerValue;
    public bool isAnsweringQuestion = true;
    public float fillFraction;
    public bool loadNextQuestion;

    [SerializeField] private Image timerImage;

    Mensagem da Unity | 0 referências
    void Update()
    {
        UpdateTime();

        if (timerImage != null)
        {
            timerImage.fillAmount = fillFraction;
        }

        if (loadNextQuestion)
        {
            // O Questionnaire vai tratar de chamar a próxima pergunta e limpar este bool
            loadNextQuestion = false;
        }
    }

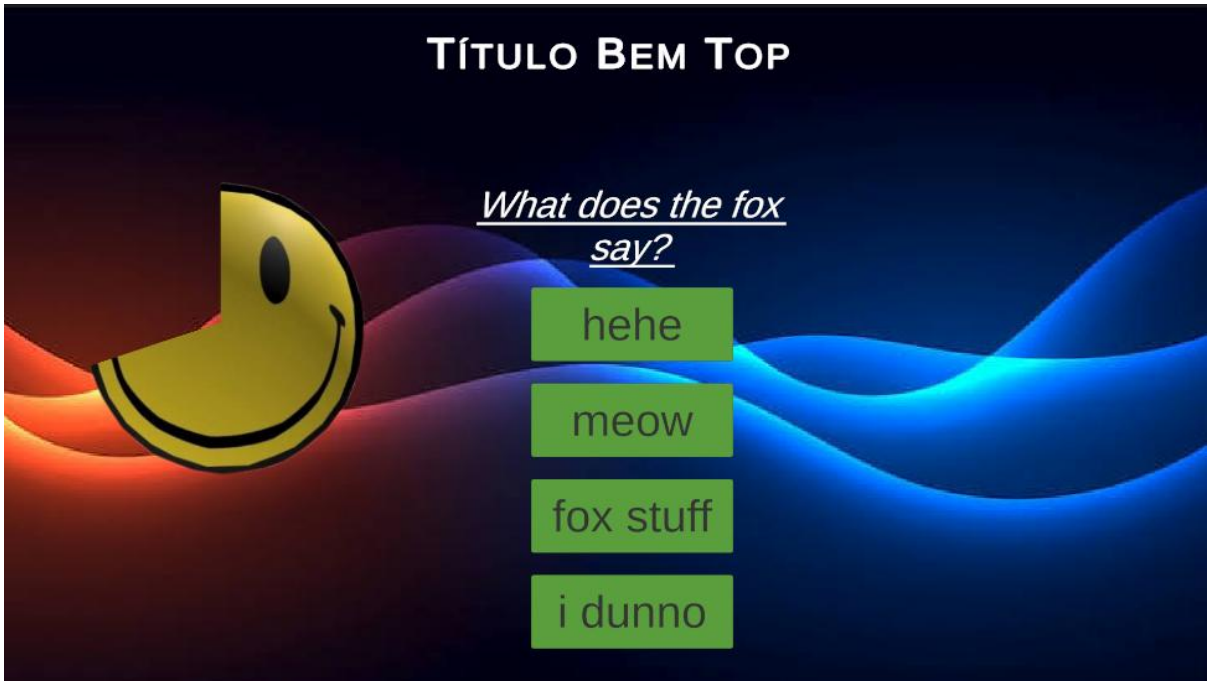
    1 referência
    public void UpdateTime()
    {
        if (isAnsweringQuestion)
        {
            timerValue -= Time.deltaTime;

            if (timerValue <= 0)
            {
                isAnsweringQuestion = false;
                timerValue = timeToShowCorrectAnswer;
            }
            else
            {
                fillFraction = timerValue / timeToCompleteQuestion;
            }
        }
        else
        {
            timerValue -= Time.deltaTime;

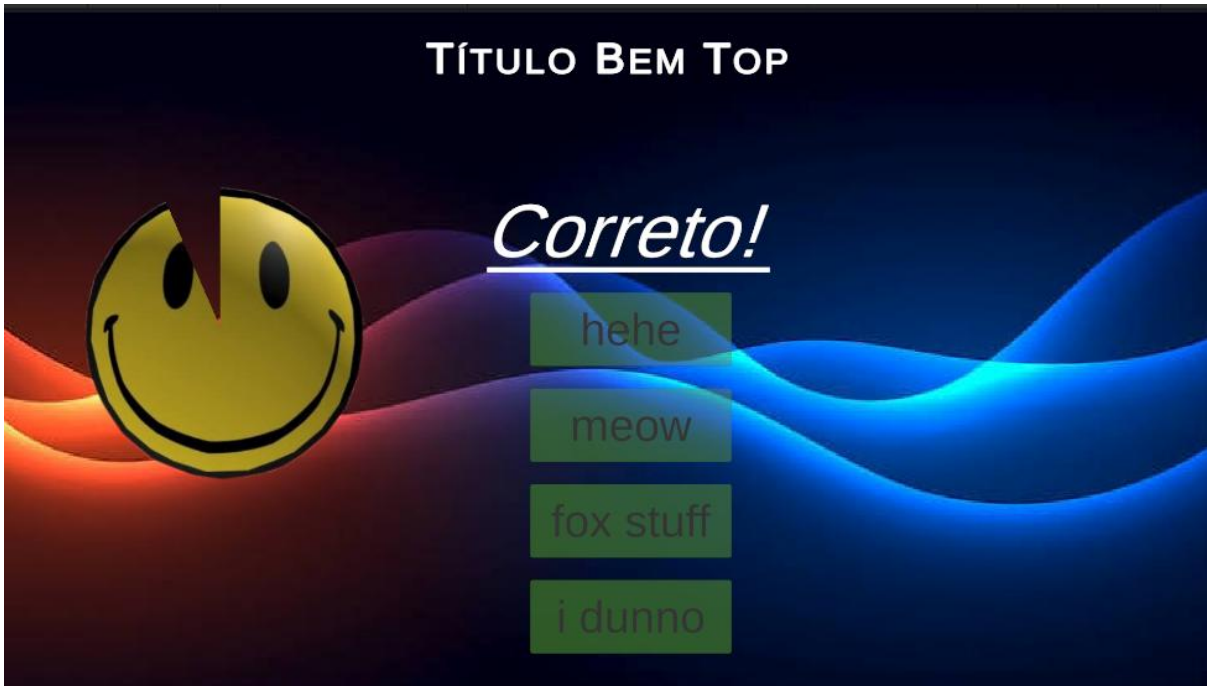
            if (timerValue <= 0)
            {
                isAnsweringQuestion = true;
                timerValue = timeToCompleteQuestion;
                loadNextQuestion = true;
            }
            else
            {
                fillFraction = timerValue / timeToShowCorrectAnswer;
            }
        }
    }

    1 referência
    public void CancelTimer()
    {
        timerValue = 0;
    }
}
```

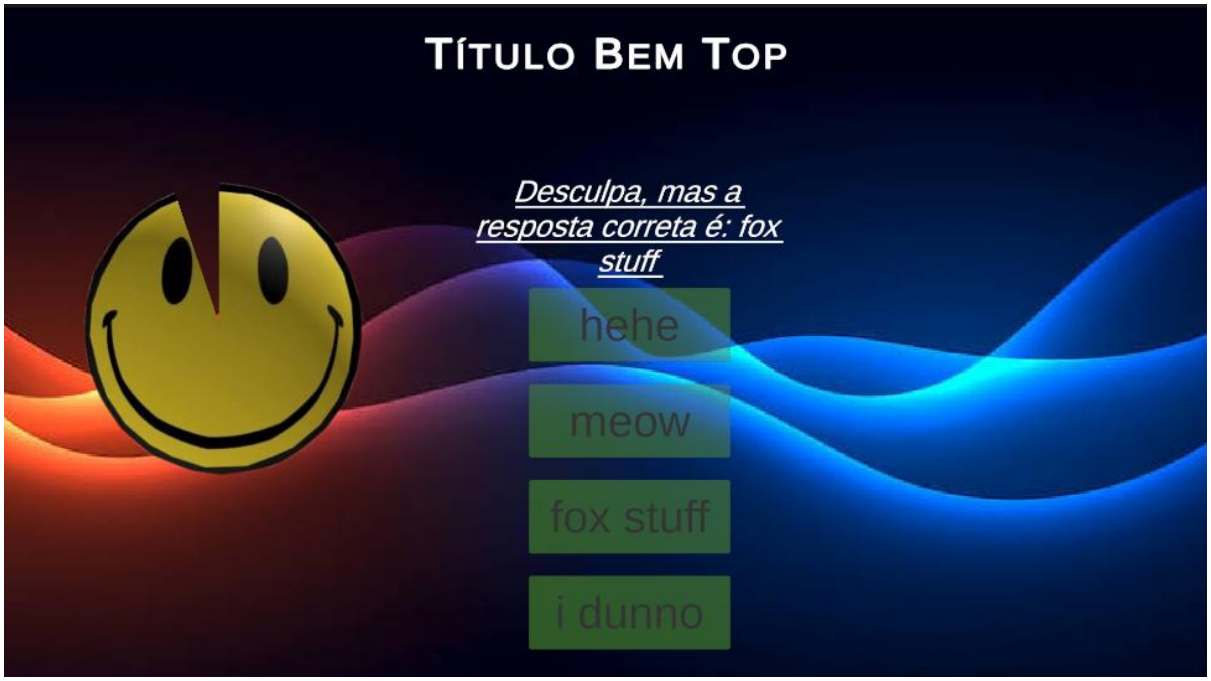
On Level 5, we coded up this timer script to run our game's pacing and get the countdown looking right. We set it up to track two phases: a 30-second window to answer the question where the image slowly disappears smoothly, and a 5-second window to show the right answer before jumping to the next question. We made a function called “CancelTimer” so we can make the clock equal to zero so it’s faster.



This is the main part where the Player has to answer the question while the ball slowly disappears counterclockwise that works as the timer.



This appears when the player gets the question right, the boxes get disabled and it shows that the player answered correctly.



This is mostly the same part, when the player answers incorrectly it shows what the correct answer is and it restarts.